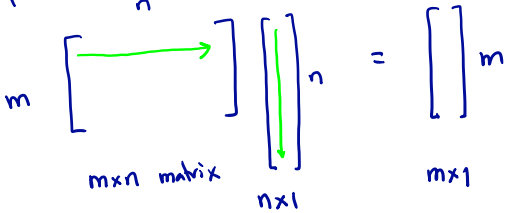
# Linear Algebra: Geometry and Algebra

Vectors & Matrices

Algebraic perspective: matrix – vector multiplication

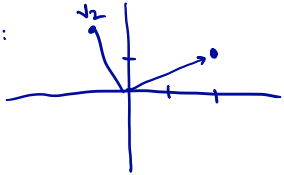



$$\overset{m\times \underline{n}}{A} \cdot \overset{\underline{n}\times k}{B} = \overset{m\times k}{C}$$

$$\overset{v_1}{\begin{pmatrix}2\\1\end{pmatrix}} \qquad \overset{A}{\begin{bmatrix}a & b\\ c & d\end{bmatrix}} : \mathbb{R}^2 \to \mathbb{R}^2$$

$2\times 2$

Geometric Perspective:




<u>Linear</u>: $A(\alpha v_1 + \beta v_2) = \alpha A v_1 + \beta A v_2$.

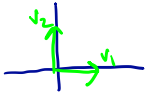
# Basic Geometric Concepts

## Linear Independence

$v_1, v_2, \ldots, v_n$ are called <u>linearly independent</u>

if: $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0 \Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0 \cdots = \alpha$

Ex:




$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\alpha_1 v_1 + \alpha_2 v_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{pmatrix} \alpha_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\alpha_1 = \alpha_2 = 0$

## Linear Dependence

$v_1, \ldots, v_n$ are linearly dependent

if $\exists\, \alpha_1, \ldots, \alpha_n$ <u>not</u> all zero, with

$\sum \alpha_i v_i = 0$

Ex: $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$1 \times v_1 + 1 \times v_2 + (-1) \cdot v_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

# Basic Geometric Concepts

## Basis

$\mathbb{R}^n$

$v_1, \ldots, v_n$ is called a basis for $\mathbb{R}^n$, if all $\underline{\underline{n}}$ are linearly independent.

# Matrices and Geometry

~~Row~~ rank and ~~Column~~ rank

$$A = \begin{array}{c} \\ m \end{array} \begin{array}{c} \begin{array}{cc} w_1 & \quad n \quad w_n \end{array} \\ \begin{bmatrix} \\ \\ \\ \end{bmatrix} \end{array} \begin{array}{c} \\ v_1 \\ v_2 \\ \vdots \\ v_m \end{array}$$

Row rank: # independent rows

Column rank: # independent columns

Fact: Row Rank = Column Rank.

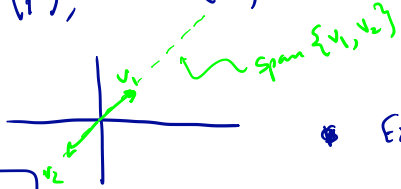
(Rank - Nullity Theorem)

# Matrices and Geometry

## Span and Range

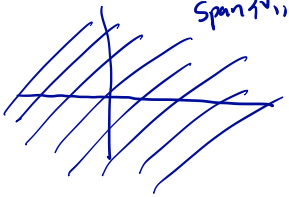
$v_1, \dots, v_n$ : span $= \{ z = \sum \alpha_i v_i : \alpha_i \in \mathbb{R} \}$

Ex: $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$




Ex: $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

span$\{v_1, v_2\} = \mathbb{R}^2$.



Range$(A) = \{ z = Av,$ some $v \}$

$$A = \begin{bmatrix} | & | & & | \\ w_1 & w_2 & \cdots & w_n \\ | & | & & | \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \Rightarrow Av = \sum v_i \begin{bmatrix} | \\ w_i \\ | \end{bmatrix}$$

Range$(A)$ = Span of Columns of $A$.

# Matrix Geometry

## Null space

$A$ is $m \times n$ matrix

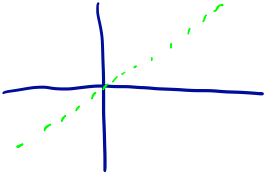
Null Space of $A$ = Null$(A)$

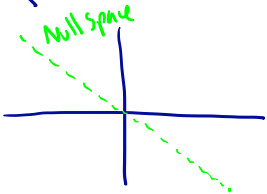
$$= \{ v \in \mathbb{R}^n : Av = 0 \}$$

Ex: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Null Space $(A) = \left\{ \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}$

Range$(A)$ = span$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ =





# Some Basic Definitions

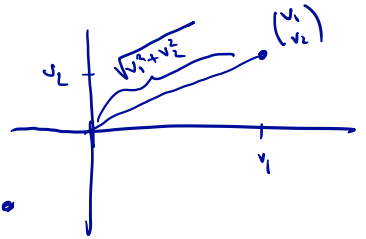
Dot product : $v, u \in \mathbb{R}^n$

$$v^T u = v \cdot u = \langle v, u \rangle = \sum_{i=1}^{n} v_i u_i$$

$v^T u = u^T v$ since: $\sum u_i v_i = \sum v_i u_i$

L p norms, and Euclidean norm

Euclidean norm aka the 2-norm,




$v \in \mathbb{R}^n$ $\quad \|v\|_2 = \sqrt{\sum v_i^2}$

$1 \leq p \leq \infty$

p-norm ($L^p$-norm)

$$\|v\|_p = \left(\sum |v_i|^p\right)^{1/p}$$

$$\|v\|_\infty = \max_i |v_i|$$

# Some Basic Definitions

## Cauchy Schwarz

$u, v \in \mathbb{R}^n$ $\quad u^T v \leq \|u\|_2 \cdot \|v\|_2$

equality iff $u$ and $v$ are pointing in same dir.

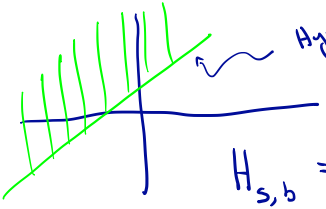
Ex: $u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ / Ex: $u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

## Hölder's Inequality

$$u^T v \leq \|u\|_p \cdot \|v\|_q, \quad \text{where } \frac{1}{p} + \frac{1}{q} = 1$$

Check that C-S is a special case

# Half spaces

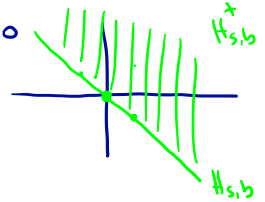


Hyperplane :

$$H_{s,b} = \{ x \in \mathbb{R}^n : s^T x = b \}$$

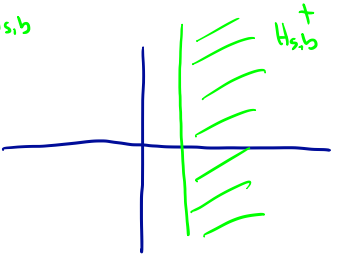
$$H^+_{s,b} = \{ x \in \mathbb{R}^n : s^T x \geq b \}$$

Ex : $s = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $b = 0$




$s^T x = 0$ : $x_1 + x_2 = 0$

Ex : $s = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $b = 1$




$s^T x = 1$ : $x_1 + 0 x_2 = 1$

$x_1 = 1$